

24-May-2026

Agenda:

ML-1

- why ML exist
- AI vs ML vs DL
- types of ML
- Data, features, Data splitting
- Discussion

Machine learning

→ How does youtube recommend videos?

based on browsing history

→ How does gmail detect spam?

```
def [ ]: email body
    if [ ]:
        return spam
    else:
        return not spam
```

def []

==

→ for each complex topic/problem, we can have million of values.

Q. Rain prediction system

cloud %.

humidity %.

wind speed

→ Rain

→ Not rain

```
if cloud > 70%
AND humidity > 80%
AND wind speed < 20
Then rain
```

humidity : 75% , windspeed is 22 , cloud 60%

problem: rule become huge, difficult, impossible to maintain

Instead of writing rule manually, we allow computer to observe data, learn patterns & improve automatically: Machine Learning

layman analogy:-

teach a child:

what is a dog? ← goal

if 4 legs
And tail
And fur
Then dog

show examples

this is a dog

this is a dog

this is not a dog

↓
child learns

ML works similarly from pattern.

AI → Artificial Intelligence

→ AI is the broader field of making machines perform tasks that normally require human intelligence.

Not an AI → AI does not mean emotion or self-awareness.

human intelligence:

(RL)

→ speaking
→ understanding language
(NLP)

→ driving
→ recognize faces
(computer vision)

→ decision making
(Agentic AI)

Application

Face unlock

Translate

Alexa

self driving car

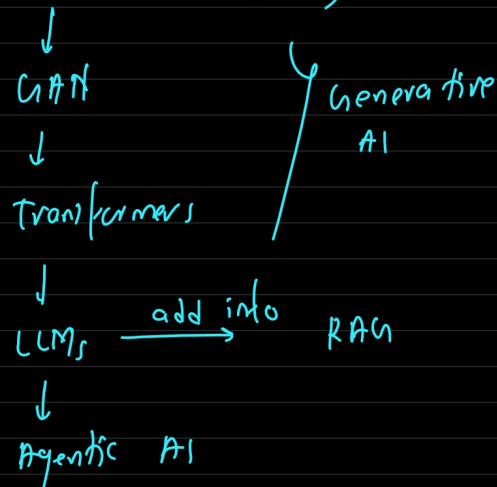
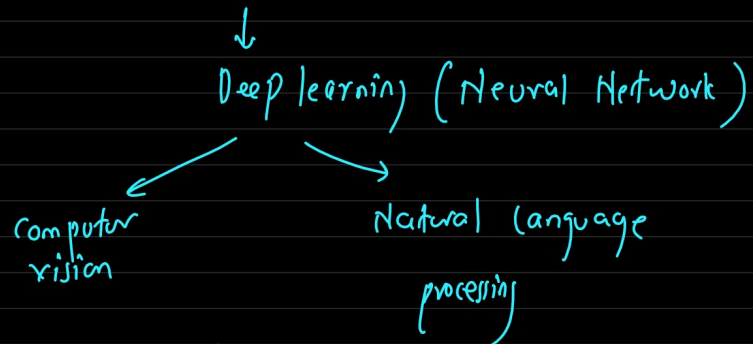
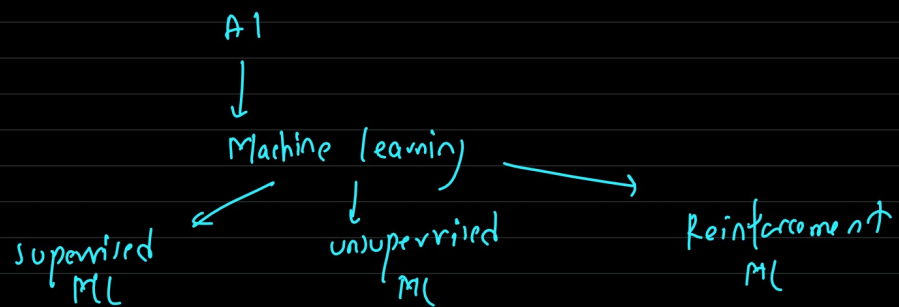
AI Task

Face Recognition

language understanding

speech understanding

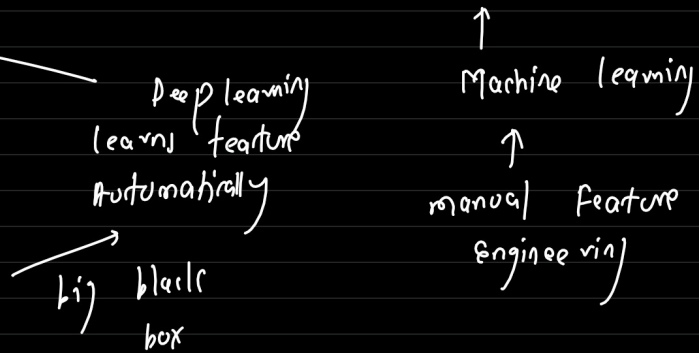
Decision Making



Real-world : Example

Experience	Salary
1	30k
2	40k
5	75k

→ pattern → relationship between experience & salary



why DL become popular:

- massive data
- computation, GPU
- internet-scale

computer vision:

- self-driving car
- face recognition
- medical imaging

NLP:

- chatbot
- translation
- summarization
- chatbot

Types of Machine Learning

How you are learning datascience?

- from mentors
- observation
- from mistakes

mentor gives:

question / topic
answers / explanation

supervised learning → labelled data → Features → X
Target → y

Examples

Examples	Target/output
House price prediction	→ price
Spam detection	→ spam/not spam
Disease prediction	→ Disease type
Student marks prediction	→ marks

supervised
ML (X, y)

Regression $\rightarrow$ output is continuous [salary, temp, stock price]

classification $\rightarrow$ output categories [spam/not spam, cat/dog, fraud/not-fraud]

predict age $\rightarrow$ Reg

predict gender $\rightarrow$ cl

predict salary $\rightarrow$ Reg

predict pass/fail $\rightarrow$ cl

Supermarket:

data: age, spending, frequency, location
 X $\rightarrow$ rich customers
discount hunters
frequent buyers
spending behaviour

X (pattern) $\rightarrow$ groups (hidden)

$\rightarrow$ answer not available

unsupervised learning

examples:

$\rightarrow$ customer segmentation

$\rightarrow$ anomaly detection (X) $\rightarrow$ transactions $\rightarrow$ ML $\rightarrow$ outlier
Huge

$\rightarrow$ recommendation grouping $\rightarrow$

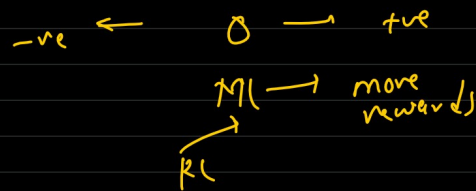


Reinforcement learning:

- rewards
- punishment
- total & error



→ dog learns behavior



Game - Mario :-

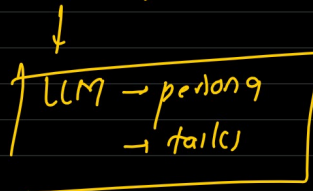
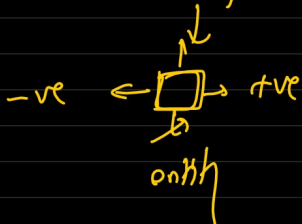
Initially: dies repeatedly

eventually:

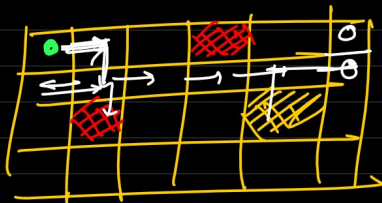
- learn jumps
- obstacles
- strategies

through reward

RL Agents $\neq$ Agentic AI Agents



$+1$ ~~$+1$~~ $+1$ → 0 → $+1$ $+1$ ~~$+1$~~

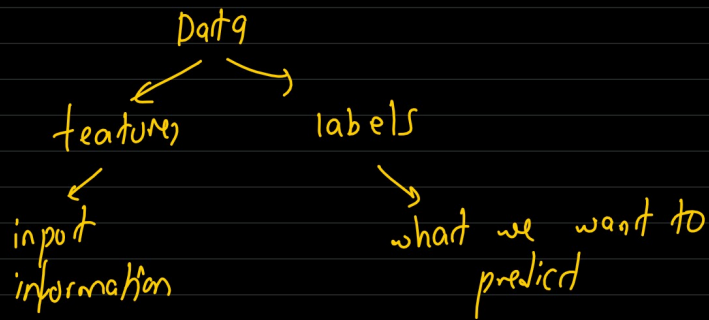


RL → Chess game →

free →
not →

- Netflix recommendation → supervised
- spam detection → supervised / unsupervised
- robot learning to walk → reinforcement
- customer clustering → unsupervised

Data understanding



Data splitting : —

Teacher: pattern (100) → ML (90/100) → 100%

→ Exact exam question before exam.

10/10 (Training is still required)

→ score: 100%.

→ memorize data/answer

→ Not learn pattern



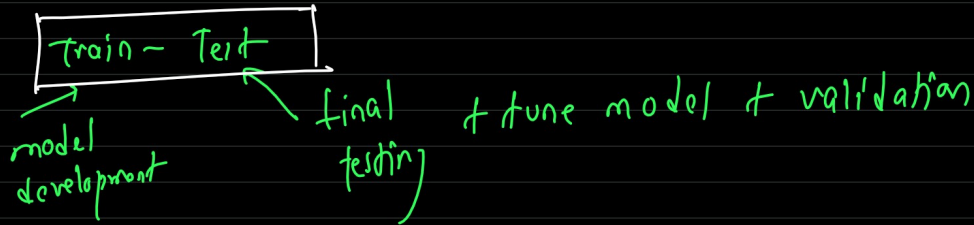
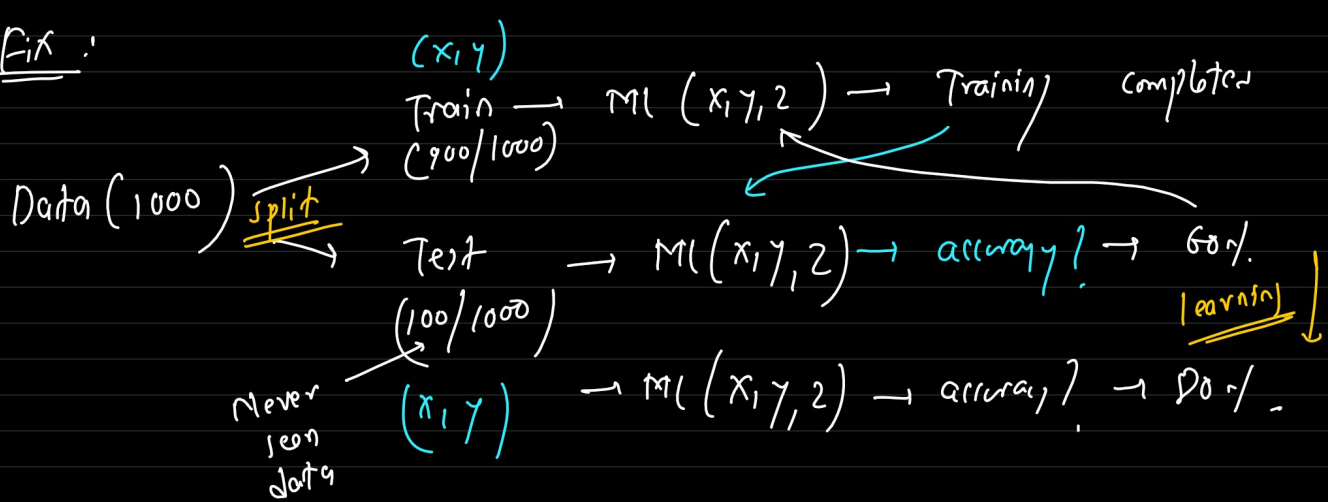
1000 → ML (Trained)

1000 → ML → 100%

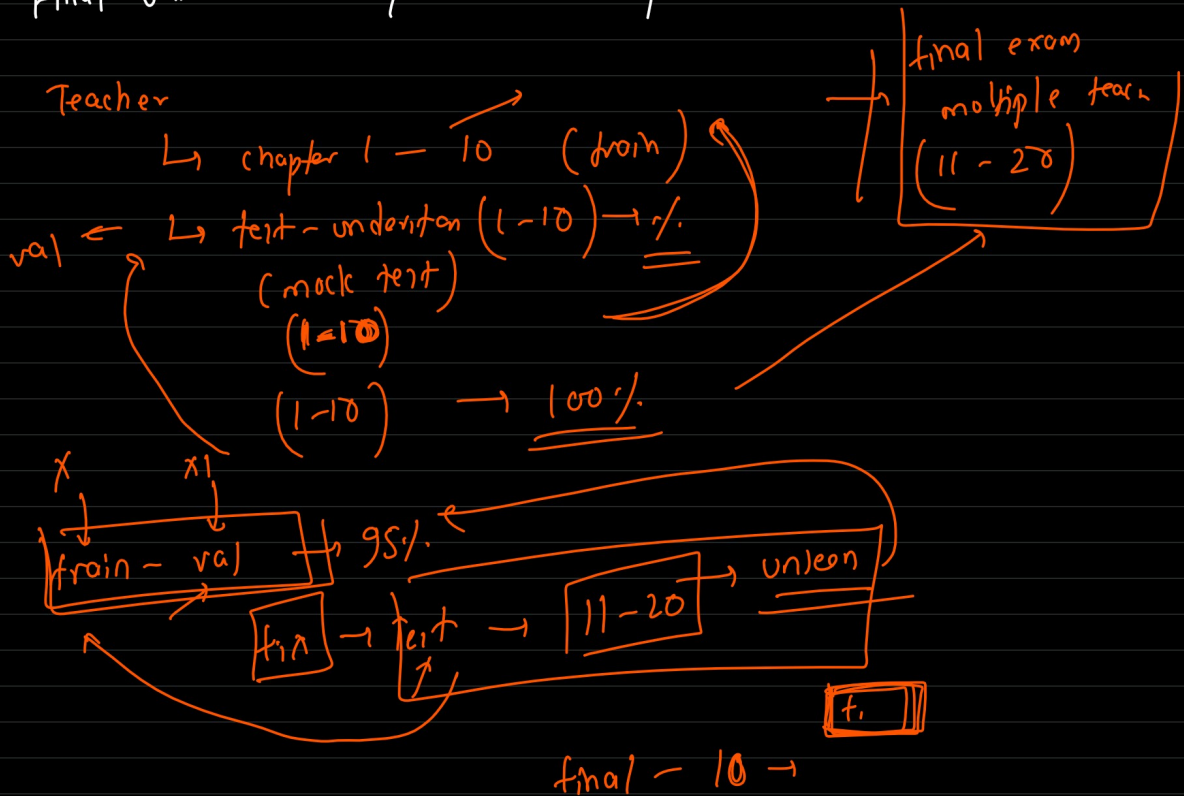
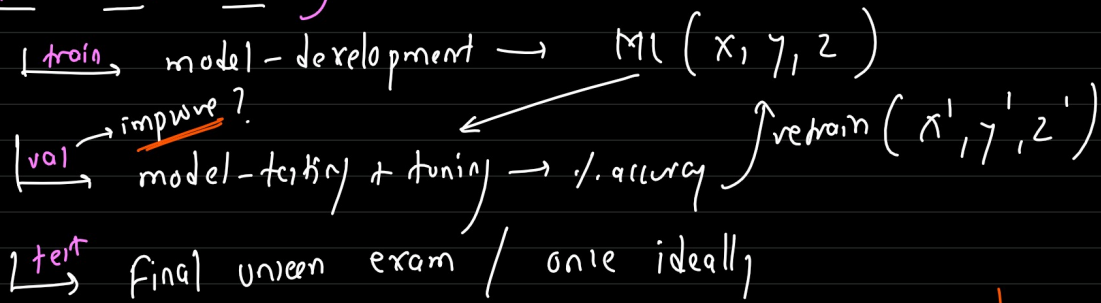
1 → ML → 0%

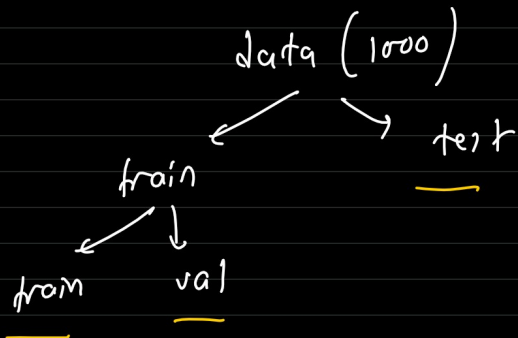
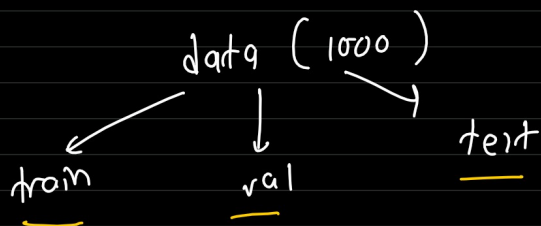
→ memory / no-learning

Fix:



Train-val-Test





data (1000 rows)

train	→ highest count	→ 90%	→ 900
val	→	→ 5%	→ 50
test	→	→ 5%	→ 50

100 (data) → ML → 90 : 5 : 5
 1000 (data) → ML → 70 : 15 : 15
 1000 (data) → ML → 80 : 10 : 10

Business : — (split → ML, AI)

